

L'Hôpital's Rule: From Calculus to Real Analysis

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1 An elementary calculus proof

Suppose that f and g are continuous on an open interval containing a real number a , differentiable there except possibly at a , and satisfy $f(a) = g(a) = 0$. Assume that $g'(x) \neq 0$ whenever $x \neq a$ is sufficiently close to a , and that $f'(x)/g'(x)$ tends to a finite real number L as $x \rightarrow a$. We want to prove that $f(x)/g(x)$ also tends to L .

The bridge between function values and derivatives is Cauchy's Mean Value Theorem. Take $x \neq a$ sufficiently close to a . First, $g(x)$ cannot be zero: otherwise $g(a) = g(x) = 0$, and Rolle's theorem would produce a point strictly between a and x where g' vanishes. Cauchy's Mean Value Theorem therefore gives a point c_x strictly between a and x such that

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c_x)}{g'(c_x)}.$$

As $x \rightarrow a$, the intermediate point c_x also approaches a , because it lies between a and x . The hypothesis on the derivative ratio then implies that the right-hand side tends to L . Hence $\lim_{x \rightarrow a} f(x)/g(x) = L$.

This proof contains the essential idea of the finite $0/0$ form of L'Hôpital's rule: a ratio of function increments is exactly a ratio of derivatives at an intermediate point, and that point is forced toward the limiting point. The notation $0/0$ describes the separate limits of the numerator and denominator; no step of the proof divides zero by zero.

The argument is already valid under the assumptions just stated. A more explicit real-analysis proof will clarify the limiting argument and also remove an unnecessary restriction: the original functions need not be defined or continuous at a .

2 What the elementary proof assumes

Continuity at a , together with $f(a) = g(a) = 0$, implies that both functions tend to zero at a . But the converse requires care. Knowing $\lim_{x \rightarrow a} f(x) = 0$ does not tell us that $f(a)$ exists, or that it equals zero if it does exist. A limit at a concerns values at points near a other than a itself.

For example, define $f(x) = x$ for $x \neq 0$ and $f(0) = 1729$. Then $\lim_{x \rightarrow 0} f(x) = 0$, although f is not continuous at zero. Alternatively, we could leave $f(0)$ undefined, and the same limit would still hold. Since L'Hôpital's rule concerns a limit of a quotient, its natural hypotheses should not depend on such an irrelevant endpoint value.

Nevertheless, the elementary proof uses a as an endpoint of a closed interval. Cauchy's Mean Value Theorem requires continuity on that closed interval, including its endpoints. This creates an apparent difficulty: how can we use the theorem when the original functions may not even

be defined at a ? The answer is to construct continuous extensions, apply the theorem to those extensions, and transfer the conclusion back to the original functions. Each part of that transfer must be justified.

3 Continuous extension is a legitimate reduction

Suppose f is defined on a punctured neighborhood of a and $\lim_{x \rightarrow a} f(x) = 0$. Define a new function by

$$F(x) = \begin{cases} f(x), & x \neq a, \\ 0, & x = a. \end{cases}$$

Then F is continuous at a . Indeed, for every $\varepsilon > 0$, the assumed limit supplies a $\delta > 0$ such that $0 < |x - a| < \delta$ implies $|f(x)| < \varepsilon$. If $|x - a| < \delta$ and $x \neq a$, this gives $|F(x) - F(a)| < \varepsilon$; if $x = a$, the same inequality holds because its left-hand side is zero. These two cases establish continuity at a directly from its definition.

This construction does not assume the conclusion of L'Hôpital's rule. It uses only the already known limit of f itself. Nor does it assert that the original f was continuous at a . The functions f and F may have different domains or different values at a , and that distinction remains in force throughout the argument.

There are two reasons why replacing f by F is useful here. First, $F(x) = f(x)$ at every nearby point other than a , so their punctured limits agree. More generally, two functions that agree throughout a punctured neighborhood of a have exactly the same limit behavior at a . This follows immediately from the definition of limit, whose hypotheses only test points with $0 < |x - a| < \delta$.

Second, derivatives away from a are preserved. For any fixed $t \neq a$ in the domain, there is a sufficiently small open interval around t that does not contain a . The functions F and f agree throughout that interval, so their difference quotients at t agree for all sufficiently small nonzero increments. Consequently $F'(t) = f'(t)$. Agreement merely at the single point t would not establish equality of derivatives; agreement on a neighborhood does.

Construct G from g in the same way, assigning $G(a) = 0$. Wherever the original quotient is defined near a , we have $F(x)/G(x) = f(x)/g(x)$. Likewise, $F'(t)/G'(t) = f'(t)/g'(t)$ for $t \neq a$ whenever the denominators are nonzero. Thus the extensions provide the missing endpoint continuity while preserving the quantities used in the proof.

Nothing here guarantees differentiability at a , and nothing requires it. For example, the continuous extension of $f(x) = |x|$ from $\mathbb{R} \setminus \{0\}$ is not differentiable at zero. In the mean value theorem argument, a is an endpoint; differentiability is required only in the open interval between the endpoints.

Calling this a continuous extension across a removable discontinuity is customary when the endpoint value is absent or incorrect. The extension is possible precisely because the prescribed finite limit already exists. Changing one value cannot repair the oscillation of $\sin(1/(x - a))$ at a , nor can it produce a finite continuous extension of $1/(x - a)$ there.

4 A precise real-analysis theorem

We now state the result entirely on a punctured interval. This formulation avoids making any assumptions about endpoint values and specifies where all the derivatives and quotients are defined.

Theorem 4.1 (L'Hôpital's rule, finite 0/0 form). *Let $a \in \mathbb{R}$, let $r > 0$, and set $D = (a - r, a + r) \setminus \{a\}$. Let $f, g : D \rightarrow \mathbb{R}$ be differentiable, and suppose $g'(x) \neq 0$ for every $x \in D$. Assume that*

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0, \quad \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L \in \mathbb{R}.$$

Then $g(x) \neq 0$ for every $x \in D$, and $\lim_{x \rightarrow a} f(x)/g(x) = L$.

Requiring g' to be nonzero on this whole punctured interval entails no loss relative to requiring it only sufficiently near a : in that case, we first shrink r . Neither continuity of the derivatives nor differentiability at a is assumed.

Proof. Define F and G on $(a - r, a + r)$ by retaining the values of f and g on D and setting $F(a) = G(a) = 0$. The assumed limits imply that both extensions are continuous at a . On D , they are differentiable, with $F' = f'$ and $G' = g'$, and are therefore continuous there as well.

For $x \in D$, let $J_x = [\min\{a, x\}, \max\{a, x\}]$. The functions F and G are continuous on J_x and differentiable on its interior. If $g(x) = 0$, then $G(x) = G(a) = 0$, so Rolle's theorem supplies an interior point ξ with $G'(\xi) = 0$. But $\xi \in D$ and $G'(\xi) = g'(\xi) \neq 0$, a contradiction. Thus $g(x) \neq 0$ for every $x \in D$.

Let $\varepsilon > 0$ be arbitrary. By the assumed limit of the derivative ratio, there exists $\eta > 0$ such that, for every $t \in D$,

$$0 < |t - a| < \eta \implies \left| \frac{f'(t)}{g'(t)} - L \right| < \varepsilon.$$

Put $\delta = \min\{\eta, r\}$, and fix any real x satisfying $0 < |x - a| < \delta$. In particular, $x \in D$. Cauchy's Mean Value Theorem, applied to F and G on J_x , gives a point c in the interior of J_x for which

$$(F(x) - F(a))G'(c) = (G(x) - G(a))F'(c).$$

This identity has the same orientation for both $x > a$ and $x < a$: reversing the endpoints changes the sign of both increments. Substituting the definitions of the extensions gives $f(x)g'(c) = g(x)f'(c)$. The factors $g(x)$ and $g'(c)$ are nonzero, by the preceding argument and the hypothesis, respectively. Division is therefore legitimate and yields $f(x)/g(x) = f'(c)/g'(c)$.

Since c lies strictly between a and x , it satisfies $0 < |c - a| < |x - a| < \delta \leq \eta$. Applying the derivative-ratio estimate at $t = c$ now gives

$$\left| \frac{f(x)}{g(x)} - L \right| = \left| \frac{f'(c)}{g'(c)} - L \right| < \varepsilon.$$

The choice of δ depends on ε , not on x . Since x was arbitrary subject to $0 < |x - a| < \delta$, this proves the required limit by its ε - δ definition. $\square$

5 What changed between the proofs?

The mathematical mechanism did not change. Both proofs use Cauchy's Mean Value Theorem to represent a quotient of increments by a quotient of derivatives at one intermediate point. The second proof separates the steps that make this representation legitimate and explicitly shows how the limit estimate is transferred.

The first change concerns the statement being proved. The elementary version assumes continuity at a and zero endpoint values. The second version assumes only zero limits and differentiability away from a . It therefore covers functions whose endpoint values are undefined or arbitrary.

Continuous extension reduces this more general situation to the setting of the elementary proof. This reduction could equally well be included in an introductory calculus proof; it is not a technique reserved for advanced analysis.

The second change concerns division. Cauchy's Mean Value Theorem initially gives a cross-multiplied identity. The quotient form follows only after checking the relevant denominators. The condition $g'(x) \neq 0$ supplies one denominator directly, while Rolle's theorem, combined with the zero limit of g , supplies the other. The zero-limit condition matters here: a function can have a nowhere-zero derivative and still vanish somewhere, as $g(x) = x - 1$ illustrates.

The third change concerns the intermediate point. Writing $c_x \rightarrow a$ is a concise way to express a valid argument, but its justification is the inequality $0 < |c_x - a| < |x - a|$. The point generally depends on x , and Cauchy's theorem need not determine it uniquely. We neither assume nor need continuity, measurability, or an explicit formula for a selection $x \mapsto c_x$.

In fact, the detailed proof does not construct such a function at all. It fixes an arbitrary x , obtains one suitable c , and applies an estimate that holds for every point sufficiently close to a . The quantifier order is essential: first choose δ from ε , then consider any eligible x , and only then use the existence of c . The same δ works for every such x because every possible intermediate point lies inside the required neighborhood.

It would be misleading to identify rigor with educational level or with the number of displayed quantifiers. This is an elementary real-analysis theorem with a fully rigorous proof; no specifically graduate machinery is needed. An experienced mathematician might compress the second proof into a paragraph, leaving standard deductions implicit. What matters is that the deductions are justified, not that every proof spells out each one at maximum length.

6 Why the common intermediate point matters

Applying the ordinary Mean Value Theorem separately to F and G would give points ξ_x and ζ_x between a and x such that $f(x) = f'(\xi_x)(x - a)$ and $g(x) = g'(\zeta_x)(x - a)$. After division, this would yield $f(x)/g(x) = f'(\xi_x)/g'(\zeta_x)$. Both intermediate points approach a , but the hypothesis controls $f'(t)/g'(t)$ at a *common* argument. It does not in general control ratios evaluated at two different arguments.

For a concrete illustration, take $f(t) = g(t) = t^2$ and $a = 0$. The same-argument derivative ratio is identically one on the punctured domain. Yet for positive x , the two points $u_x = x/2$ and $v_x = x/3$ both lie between zero and x , while $f'(u_x)/g'(v_x) = 3/2$. These points are not asserted to be the mean value points for those particular increments; the example shows exactly why their location and convergence alone cannot justify the desired inference.

Cauchy's theorem supplies the additional relation that the separate applications do not: one point serves both derivatives. Its exact identity is stronger than a heuristic assertion that sufficiently small increments are approximately derivatives times the same displacement. The latter heuristic needs further error control, especially when the denominator derivative approaches zero.

7 Scope and logical direction

The theorem proved here concerns a finite limiting point, zero limits for both functions, and a finite limit of the derivative ratio. The same endpoint argument gives one-sided versions by working only on the appropriate side of a . Infinite derivative-ratio limits can be treated by replacing the ε estimate with the defining threshold inequalities for divergence to infinity. Versions involving unbounded denominators or a limiting point at infinity require additional

arguments; they are not established merely by the continuous-extension proof above.

L'Hôpital's rule is a sufficient criterion, not an equivalence. Even in the finite $0/0$ setting, the quotient may converge while the derivative ratio does not. For $x \neq 0$, take $f(x) = x + x^2 \sin(1/x)$ and $g(x) = x$. Both tend to zero and $g'(x) = 1$, while

$$\frac{f(x)}{g(x)} = 1 + x \sin(1/x) \longrightarrow 1, \quad \frac{f'(x)}{g'(x)} = 1 + 2x \sin(1/x) - \cos(1/x).$$

The derivative ratio equals zero along $x_n = 1/(2\pi n)$ and two along $y_n = 1/((2n+1)\pi)$, for positive integers n , so it has no limit at zero. Failure of the derivative-ratio hypothesis therefore leaves the original limit undecided.

The central lesson of the proof is the relationship between local information and a limit. We know how the derivative ratio behaves at every sufficiently nearby punctured point. Cauchy's theorem identifies the original quotient with that ratio at one such point. Continuous extension makes the theorem applicable at the endpoint without altering either the original quotient nearby or the derivatives used to estimate it. Every modification serves a stated hypothesis, and every conclusion about an auxiliary function is transferred through an exact equality.